

GC University Lahore

BS Computer Science Examination: Final 2024

Semester: 7th

Course Code: CS-4101

Title: Principles of Programming Languages

Time Allowed: 155 minutes

Max. Marks: 40

Subjective Type

Attempt any Four Questions. All questions carry equal marks (10 Marks each).

Question 2 (Marks 10)

2(a) Convert the following grammar from BNF to EBNF: (5 Marks)

```

    <expr> ::= <integer_expr> | <boolean_expr>
    <integer_expr> ::= <term>
                        | <integer_expr> "+" <term>
                        | <integer_expr> "-" <term>
    <term> ::= <element>
              | <term> "*" <element>
              | <term> "/" <element>
    <element> ::= <numeral>
                | <identifier>
                | "(" <integer_expr> ")"
                | "-" <element>
    <boolean_expr> ::= <boolean_term>
                    | <boolean_expr> "or" <boolean_term>
    <boolean_term> ::= <boolean_element>
                    | <boolean_term> "and" <boolean_element>
    <boolean_element> ::= "true"
                        | "false"
                        | <identifier>
                        | <comparison>
                        | "not" " ( " <boolean_expr> " ) "
```

$$\begin{aligned}
 & | " (" \langle \text{boolean_expr} \rangle ") " \\
 \langle \text{comparison} \rangle & ::= \langle \text{integer_expr} \rangle "<=" \langle \text{integer_expr} \rangle \\
 & | \langle \text{integer_expr} \rangle "<" \langle \text{integer_expr} \rangle \\
 & | \langle \text{integer_expr} \rangle "=" \langle \text{integer_expr} \rangle \\
 & | \langle \text{integer_expr} \rangle ">=" \langle \text{integer_expr} \rangle | \\
 & \langle \text{integer_expr} \rangle ">" \langle \text{integer_expr} \rangle \\
 & | \langle \text{integer_expr} \rangle "<>" \langle \text{integer_expr} \rangle
 \end{aligned}$$

2(b) Compute the weakest precondition for each of the following assignment statements and post-conditions: (5 Marks)

- $a = 2 * (b - 1) - 1 \{a > 0\} \ x = 2$
- $* y + x - 1 \{x > 11\}$

Question 3 (Marks 10)

3(a) Prove that the following grammar is ambiguous: (3 Marks)

$$\begin{aligned}
 \langle S \rangle & \rightarrow \langle A \rangle \\
 \langle A \rangle & \rightarrow \langle A \rangle "+" \langle A \rangle | \text{id} \\
 \langle \text{id} \rangle & \rightarrow a | b | c
 \end{aligned}$$

3(b) Consider the following given grammar: (7 Marks)

$$\begin{aligned}
 \text{Number} & \rightarrow \text{Sign List Sign} \\
 & \rightarrow + | - \\
 \text{List} & \rightarrow \text{Bit List} \\
 \text{Bit} & \rightarrow 0 \\
 \text{Bit} & \rightarrow 1
 \end{aligned}$$

- **a)** Derive $\rightarrow -1011$
- **b)** Describe semantic rules to calculate its decimal value.
- **c)** Draw a decorated parse tree (for the decimal value derived above).

Question 4 (Marks 10)

4(a) Briefly explain the following with examples: (5 Marks)

1. Denotational Semantics
2. Axiomatic Semantics

4(b) Consider the following JAVA code. (i) Mention its binding type, and (ii) Write the output: (5 Marks)

```

class addition {
    public int add(int a, int b)
    {
        return a + b;
    }

    public int add(int a, int b, int c) {
        return a + b + c;
    }
}

public class binding {
    public static void main(String args[])
    {
        addition a1 = new addition();
        int a, b, c;
        a = 10;
        b = 20;
        c = 30;
        System.out.println(a1.add(a, b));
        System.out.println(a1.add(a, b, c));
    }
}

```

+-----+	
Binding Type:	
Output:	
+-----+	

Question 5 (Marks 10)

$\langle \text{assign} \rangle \rightarrow \langle \text{id} \rangle "=" \langle \text{expr} \rangle$
 $\langle \text{id} \rangle \rightarrow A \mid B \mid C$
 $\langle \text{expr} \rangle \rightarrow \langle \text{id} \rangle "*" \langle \text{expr} \rangle \mid \langle \text{id} \rangle "+" \langle \text{expr} \rangle \mid "(" \langle \text{expr} \rangle ")" \mid \langle \text{id} \rangle$

Using the above grammar, show a parse tree and a leftmost derivation for each of the following statements:

- i) $A = A * (B + (C * A))$ ii) $B =$
- $C * (A * C + B)$ iii) $A = A * (B$
- $+ (C))$

Question 6 (Marks 10)

Consider the following pseudocode, assuming nested subroutines and static scope:

- (a) What does this program print?
- (b) Show the activation record frames on the stack when subroutine A has just been called. For each frame, show the static and dynamic links.

- **(c)** Explain how subroutine A finds variable g .